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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Searching
d. Sorting

The correct answer is: Sorting

Question **2**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates
b. It complicates the analysis process
c. It makes the analysis slower
d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. Each part is sorted independently
c. The parts are merged without sorting
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It avoids the need for recursion
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Combining sorted sublists
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list
b. Arranging data in a particular format
c. Deleting data from a list
d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Insertion Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Removing duplicates from a list
b. Merging two datasets
c. Finding a specific element in a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Makes code execution slower
c. Decreases the efficiency of algorithms
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Complexity
c. Merging
d. Searching

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting is rarely used in programming
c. Sorting decreases the efficiency of selection operations
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It is less efficient than bubble sort
c. It sorts data using a single pass
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Linear Search
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120